

Attention Mechanism

ROPE was $\rightarrow \langle R_m q, R_m k \rangle = \langle q, R_{n-m} k \rangle$

The attention score b/w query at position m & key at position n depends only on relative offset $(n-m)$. This was beautiful Result.

Fast dim & Slow dim Issue $\rightarrow$

Ex $\rightarrow$ Query Vec $\rightarrow q = [q_0, q_1, q_2, \dots, q_7]$

Groups it $\Rightarrow (q_0, q_1), (q_2, q_3), (q_4, q_5), (q_6, q_7)$

$i = \{0, 1, 2, 3\}$

Rotation Speed (Freqⁿ) $\rightarrow \Omega_i = 10000^{-2i/d}$
 $d = 8$.

So we get value respectively $1.0, 0.1, 0.01, 0.001$.

Let's see what happens at a few positions:

$$\{\Omega_i = 10,000^{-2i/d}\}$$

or $\frac{1}{10,000^{2i/d}}$

Pair 0 (fastest) $\rightarrow$ Pos 0 - angle 0
($\Omega_0 = 1.0$) Pos 1 - Ang 1 Rad.

Pos 2000 Ang 2000 Rad ≈ 318 full Revolution.

Pair 3 (Slowest) $\rightarrow$ Pos 0 - Angle 0.
($\Omega_3 = 0.001$)

Pos 100 - Angle 0.1 Rad

Pos 1000 - Ang 1 Rad ($\sim 57^\circ$)

See Now, Pair 0 has already wrapped around hundreds of time by Pos 2000 whereas Pair 3 barely moved off zero

Fast dim - Distinguish nearby positions sharply (Pos 5 & 6 look very diff)

Slow dim - Distinguish far-apart positions (Pos 100 & 2000 look meaningfully diff)

Real world analogy - Clock with Multiple hands.

$\therefore$ Perplexity is fine up to L_{train}, then explodes the moment you go past it.

Technique 1 - ~~Exposed~~ Positional Interpolation →

(Extending Context Window of LLM's via Positional Interpolation)
LLAMA 1
(2048-token context Window)

Standard ROPE →

$$\text{angle} = m \cdot \theta_i \quad \begin{matrix} \uparrow \\ \text{position} \end{matrix} \quad \begin{matrix} \leftarrow \text{dim} \\ \text{pair} \end{matrix}$$

PI →

$$\text{Angle} = \frac{m}{s} \cdot \theta_i, \text{ where } s = \frac{L_{\text{new}}}{L_{\text{train}}}$$

Ex $L_{\text{train}} = 2048$

$L_{\text{new}} = 8192$

then $s = 4$.

i.e., Pos 8192 at inference produces the same rotation angle that pos 2048 produced during training. All angles stay within the training distributions

Why Interpolation & Not Extrapolation →

Naive Extrapolation

- Training saw pos.: $[0, 1, \dots, 2047]$

Inference Uses: $[0, 1, \dots, 8191]$

↑
Never seen
(Extrapolation)

[OOD Problem]

PI →

Training saw pos = $[0, 1, \dots, 2047]$

Inference Uses

= $[0, 0.25, \dots, 2047]$

all inside training Range.

(Interpolation)

∴ During training the adjacent tokens on pair 0 are separated by $0.25 \times 4 = 1.0 \text{ Rad}$
After PI with $s = 4$, adjacent tokens on pair 0 are sep^r by $0.25/4 = 0.0625 \text{ Rad}$.

Untouched Idea - RL-Transformer work.

Apply per-pathway ROPE scaling in dual pathway RSSM+Transformer WM - fast pathways (no scaling) for local dynamics, slow pathway (PI/NTK-by-parts) for long-horizon memory. Bridges Multi-Scale temporal coding to length-extrapolation literature

Drawback of NTK-Aware $\rightarrow$

Looking at the Ratio $\frac{\text{new-} \theta_i}{\text{old-} \theta_i} = S^{\frac{-2i}{d-2}}$

$i=0 \rightarrow \text{ratio} = 1 \rightarrow \text{Untouched}$

$i=1 \rightarrow \text{ratio} = S^{-1/2} \rightarrow \text{Tiny Perturbation, but non zero}$

$i = \frac{d}{2} - 1 \rightarrow \text{ratio} = S^{-1} \rightarrow \text{Full PI Scaling.}$

The Fast dim get's slightly perturbed. This is the side effect of the formula's smooth interpolation from $i=0$ to $i=d/2-1$.
- there's no Mechanism in NTK Aware that says 'Stop touching fast dims.' It just falls off gradually.

[NTK-Aware Perturbs fast dim as a side effect, not because they need scaling]

Fix $\rightarrow$

(FPDF) Final Per-dimension Freq $\rightarrow$

NTK-by-Points. (Same Block 97). $\theta_i^{\text{new}} = (1 - \gamma(r_i)) \cdot \frac{\theta_i}{S} + \gamma(r_i) \cdot \theta_i$

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Core Idea - Instead of smoothly interpolating across all dim, partition them explicitly. Leave fast dim exactly alone. Scale only the slow ones.

How to define "fast v/s slow"?

Use No. of Rotⁿ of each dim across dim training context L .

$$r_i = \frac{\theta_i \cdot L}{2\pi}$$

If r_i is large, dim completed many full rotⁿ during training. Model saw the full angle. Space $\rightarrow$ Safe.

Else $\rightarrow$ dim didn't even finish 1 Rotⁿ. Model only saw a fraction of $[0, 2\pi]$ $\rightarrow$ OOD risk when extending

3 Regions -

Region	Cnd ⁿ	Treatment
Fast	$r_i > B$ (eg. $B=32$)	Leaved one- $\theta_i^{\text{new}} = \theta_i$
Slow	$r_i < a$ (eg. $a=1$)	Full PI $\theta_i^{\text{new}} = \frac{\theta_i}{S}$
Middle	$a \leq r_i \leq B$	Smooth Blend

Ramp f^h

$$\gamma(r) = \begin{cases} 0 & \rightarrow r < a \\ 1 & \rightarrow r > B \\ \frac{r-a}{B-a} & \rightarrow \text{otherwise} \end{cases}$$

FPDF,

$\gamma=0$ (Slow) - Full PI scaling, in

$\gamma=1$ (Fast) - Completely Untouched

NTK-aware (2023) (Neural Tangent Kernel Theory) 2018

Reddy
blocc97

NTK-aware scaled RoPE allows LLAMA models to have extended (8K+) context size without any fine tuning.

Past Problems - OOD & Resolution on fast dims

PI fixes A but creates B - It applies uniform compression - every dim pays equally. The dim that didn't need fixing pay the biggest cost.

NTK Goal - Fix A without creating B.

Core Idea - Instead of scaling Pos., Scale the Bias

RoPE's frequencies are, $\phi_i = \text{base}^{-2i/d}$ ^{often,} $\{\text{base} = 10,000\}$

NTK replaces base with larger value have new base as,

$$\text{base}_{\text{new}} = \text{base} \cdot 5^{d/(d-2)}$$

$$\phi_i^{\text{new}} = \text{base}_{\text{new}}^{-2i/d}$$

Because the formula is $\exp^i$ in i , changing the base affects each dim diff[^]:

Fast dims (small i): Exponent near zero $\rightarrow \text{base}^0 = 1$ (Barely changes)

Slow dims (large i): Exponent large in magnitude $\rightarrow$ base change has huge effect $\rightarrow$ heavily compressed.

{ One No. Changes, & the exponential math distributes the effect automatically }

RLWA - The Radio Dial: An Old Analog radio has one tuning knob. The bias is "Logarithmic" - turning it a small amount barely shifts the high frequency (AM) stations packed at the top, but significantly shift the low FM sta, across the bottom.

YARN

(Yet another RoPE extension)

YARN: Efficient Context Window Extension of LLM's

- Previous Problems → (i) The Dilution Problem - (More tokens)
- Softmax denominator has more terms → Probability mass spreads thinner → the "winner" token gets a smaller share even though its logit hasn't changed.
- Positional Encoding fixes where tokens are, it doesn't fix how many are competing
- (ii) Before vs after Softmax -
- Softmax is exponential (non-linear) scaling before changes the shape of the distributions - it re-sharpens.
- Scaling after is just ~~rescaling~~ rescaling numbers that already sum to 1. The exp is the operation that translates "logit gap" into "probability gap", so you have to widen the gap before the exponential centers it.

Ex → $d = 8$ $i = 0, 1, 2, 3$
 $\text{base} = 10000$
 $L = 2048$
 $S = 4$ (extending to 8192)
 $\alpha = 1, \beta = 32$

$$Q_0 = 10000^0 = 1$$

$$\{Q_i = \text{base}^{-2i/d}\}$$

$$Q_1 = 10000^{-2/84} = 0.1$$

$$Q_2 = 10000^{-4/84} = 0.01$$

$$Q_3 = 10000^{-6/84} = 0.001$$

$$r_0 = \frac{1 \times 2048}{6.283} = 325.95 \quad \left\{ r_i = \frac{Q_i \cdot L}{2\pi} \right\} \quad \begin{matrix} \text{(Fast dim)} \\ \gamma = 1 \end{matrix}$$

$$r_1 = \frac{0.1 \times 2048}{6.283} = 32.595 \quad \text{(Fast dim)} \quad \gamma = 1$$

$$r_2 = \frac{0.01 \times 2048}{6.283} = 3.2595 \quad \text{(Middle)} \quad \begin{matrix} \alpha = 1 \\ \beta = 32 \end{matrix}$$

$$r_3 = \frac{0.001 \times 2048}{6.283} = 0.32595 \quad \text{(Slow dim)} \quad \gamma = 0$$

ramp val for Middle one (r_2).
3.2595

$$\gamma(r_2) = \frac{3.2595 - 1}{32 - 1}$$

$$\gamma(r_2) = 0.072$$

$$\gamma(r_0) = 1, \quad \gamma(r_2) = 0.072$$

$$\gamma(r_1) = 1, \quad \gamma(r_3) = 0.$$

ARN - 2 independent fixes inside ROPE -

- ① Positional fix (NTK by parts)
- ② Statistical fix (temperature) - Longer context $\rightarrow$ softmax denominator has more terms $\rightarrow$ Prob mass dilutes $\rightarrow$ Attention gets mushy. Sharpen logits before softmax to compensate.

Elegant Implementation $\rightarrow$ Both fixes live inside the ROPE module. NTK by parts modifies the frequencies (θ), temperature modifies the amplitude (cos/sin magnitudes). Attention code stays vanilla.

Maths - Freq Scaling (NTK by parts)

$$r_i = \frac{\theta_i \cdot L}{2\pi}, \quad \gamma(r_i) = \text{clip}\left(\frac{r_i - a}{\beta - a}, 0, 1\right)$$

$$\theta_i^{\text{new}} = (1 - \gamma(r_i)) \cdot \frac{\theta_i}{s} + \gamma(r_i) \cdot \theta_i$$

$\gamma = 1 \rightarrow$ Fast dim, untouched

$\gamma = 0 \rightarrow$ Slow dim, pure PI

$0 < \gamma < 1 \rightarrow$ Middle dim, blended.

Amplitude Scaling (temperature) $\rightarrow$

$$\frac{1}{t} = 0.1 \ln(s) + 1$$

$$\cos(m\theta_i) \rightarrow \sqrt{\frac{1}{t}} \cdot \cos(m\theta_i), \quad \sin(m\theta_i) \rightarrow \sqrt{\frac{1}{t}} \sin(m\theta_i)$$

why $\sqrt{1/t}$ both multiply w/ $(1/t)$ (sharpening inside Attention)